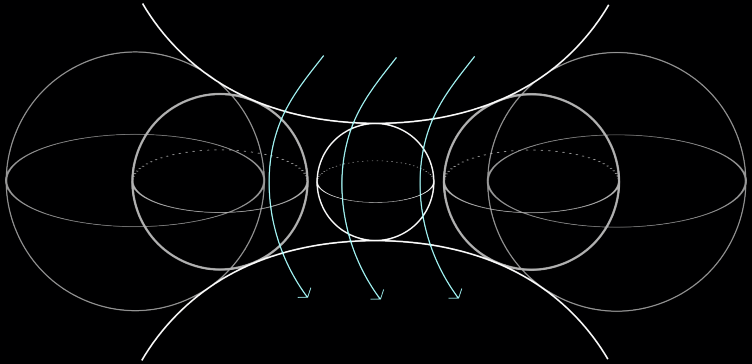


Spaces of Positive Scalar Curvature Metrics on Non-Orientable Manifolds

Milan Zerbin Göttingen, 26.11.2024



The World of Positive Scalar Curvature

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It is up to normalization uniquely determined by those properties. [Gromov 2021]

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$\rightarrow$ $\mathcal{R}^+(S^2)$ and $\mathcal{R}^+(\mathbb{R}\mathrm{P}^2)$ even contractible [Rosenberg-Stolz 2001]

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$\rightarrow$ Spinnable case by Stolz in 1992: Vanishing of $\hat{\mathfrak{a}}(M)$ neccesary and sufficient.

Uniqueness

Theorem (Ebert–Wiemeler, 2022)

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$\rightarrow$ $\mathcal{R}^+(S^n)$ has rich topology for $n \geq 6$ [Ebert–Randal-Williams 2019]

Method

Theorem (Ebert–Wiemeler, 2022)

Let M and M' be n-manifolds of tangential 2-type (B, θ) and $x \in \Omega_n^\theta$ good, such that $[M] \equiv [M'] + x$ in Ω_n^θ. Then if $n \geq 5$

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$\rightarrow \theta = \mathrm{BSpin} \rightarrow \mathrm{BO}$ gives the result on the previous slide

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$\rightarrow$ Notice, that $\left|\Omega_n^\theta / \mathcal{A}_n\right|$ is an upper bound for the amount of different homotopy types of $\mathcal{R}^+(M)$ in dimension n.

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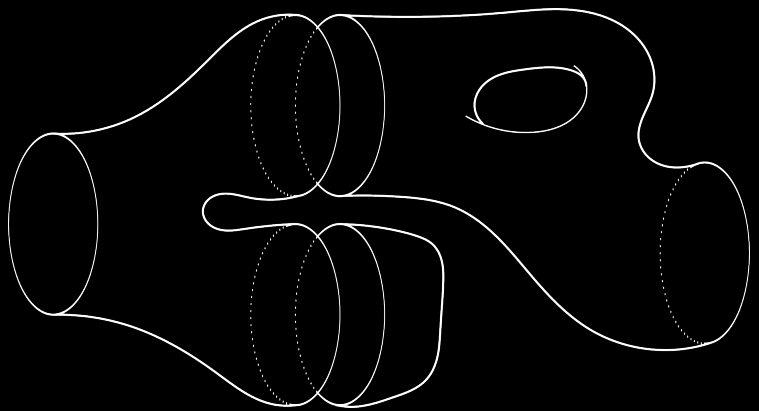
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Calculations in the Unoriented Bordism Groups

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Definition

Call two closed n-dimensional manifolds M, M' bordant, if there is a manifold W of dimension $n+1$ with boundary $\partial W = M \sqcup M'$.

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$\rightarrow$ Defines an equivalence relation on the set of all closed n-dimensional manifolds.
$\rightarrow$ The set of equivalence classes forms an abelian group $\mathfrak{N}_n$ with disjoint union $\sqcup$ as sum.

Ring Structure

Definition

The (unoriented) bordism ring is the unital, commutative, graded ring

$$\mathfrak{N}_* = \left(\bigcup_{n \in \mathbb{N}} \mathfrak{N}_n, \sqcup, \times, [\varnothing], [\{*\}] \right)$$

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$\rightarrow$ Surprisingly, $\mathfrak{N}_n$ is finite for each n.
$\rightarrow$ Furthermore, the entire ring is well understood...

Thoms work

Theorem (Thom–Dold, 1954-1956)

The unoriented bordism ring is isomorphic to a polynomial algebra over $\mathbb{Z}_2$, with precicely one generator in each degree $n \neq 2^k - 1$. Generators are represented by Dold-manifolds $P(m, n)$.

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$\rightarrow$ Define $P(m, n) = S^m \times \mathbb{C}\mathrm{P}^n/_\sim$ where $(x, z) \sim (-x, \bar{z})$.
$\rightarrow$ Notice $P(m, 0) \cong \mathbb{R}\mathrm{P}^m$.

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- In odd dimension $n = 2^r(2s+1) - 1$, we have a multiplicative generator given by $[P(2^r - 1, s2^r)]$ whenever $n \neq 2^k - 1$.

$\rightarrow$ Additive generators of $\mathfrak{N}_n$ correspond to partitions of n that contain no element of form $2^k - 1$.

Additive Generators

An additive generator of $\mathfrak{N}_n$ has the general form

$$\left[\prod_{i\in I}(\mathbb{R}\mathrm{P}^{2i})^{m_i}\times\prod_{j\in J}P(2^{r_j}-1,s_j2^{r_j})\right]$$

where $\sum_{i\in I}(2m_ii)+\sum_{j\in J}(2^{r_j}(2s_j+1)-1)=n$, and $I,J\subset\mathbb{N}$.

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$\rightarrow$ Now: Which of those are good?

Good Generators I

Lemma (Ebert–Wiemeler, 2022)

Let $M \to M'$ be a smooth fiber bundle with fiber $\mathbb{K}\mathrm{P}^{n+m-1}$. Then $[M]$ is good if either ...

- $\mathbb{K} = \mathbb{R}$, $n, m \geq 3$ with structure group $O(m) \times O(n)/\{\pm 1\}$
- $\mathbb{K} = \mathbb{C}$, $n, m \geq 2$ with structure group $U(n) \times U(m)$.
- $\mathbb{K} = \mathbb{C}$, $n + m - 1 \geq 3$ with structure group $\mathbb{Z}_2$.
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$\to$ Remember Rokhlin's Trick: $[\mathbb{R}\mathrm{P}^n \times \mathbb{R}\mathrm{P}^n] = [\mathbb{C}\mathrm{P}^n]$, $[\mathbb{C}\mathrm{P}^n \times \mathbb{C}\mathrm{P}^n] = [\mathbb{H}\mathrm{P}^n]$.

Good Generators II

Repeatedly applying the above yields, that all (additive) generators are good, except maybe those of form

$$(\mathbb{R}\mathrm{P}^2)^i \times (\mathbb{R}\mathrm{P}^4)^j \times P(1,2)^k$$

where $i = 0, 1, 2, 3$ and $j = 0, 1$ as well as $k \in \mathbb{N}$.

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Lemma (Ebert–Wiemeler)

$[P(1,2)]$ is good. Furthermore, elements of form $[M' \times P(1,2)]$ are good, if M' has a PSC-metric.

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There are only finitely many generators left of which we dont know whether they are good.

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n = 4, &\quad \mathbb{R}\mathrm{P}^4, \mathbb{C}\mathrm{P}^2 \\
n = 6, &\quad \mathbb{R}\mathrm{P}^2 \times \mathbb{C}\mathrm{P}^2, \mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^4 \\
n = 8, &\quad \mathbb{R}\mathrm{P}^4 \times \mathbb{C}\mathrm{P}^2 \\
n = 10, &\quad \mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^4 \times \mathbb{C}\mathrm{P}^2
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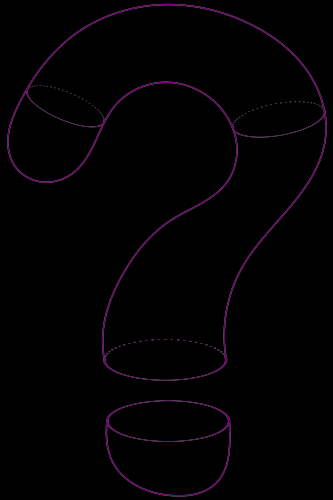
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\begin{aligned}
n = 2, &\quad \mathbb{RP}^2 && \leftarrow \text{Theorem doesn't apply} \\
n = 4, &\quad \mathbb{RP}^4, \mathbb{CP}^2 && \leftarrow \text{Theorem doesn't apply} \\
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\end{aligned}
$$

Good Generators III

There are only finitely many generators left of which we dont know whether they are good.

$$
\begin{aligned}
n = 2, &\quad \mathbb{R}P^2 && \leftarrow \text{Theorem doesn't apply} \\
n = 4, &\quad \mathbb{R}P^4, \mathbb{C}P^2 && \leftarrow \text{Theorem doesn't apply} \\
n = 6, &\quad \mathbb{R}P^2 \times \mathbb{C}P^2, \mathbb{R}P^2 \times \mathbb{R}P^4 \\
n = 8, &\quad \mathbb{R}P^4 \times \mathbb{C}P^2 \\
n = 10, &\quad \mathbb{R}P^2 \times \mathbb{R}P^4 \times \mathbb{C}P^2
\end{aligned}
$$

$\rightarrow$ In particular, for dimension > 10 all additive generators are good, and we obtain our theorem.



Main Result

Theorem

Let M be a n-dimensional Manifold of tangential 2-type $\mathrm{id}_{\mathrm{BO}}$, and $n > 10$. Then

$$\mathcal{R}^+(M) \simeq \mathcal{R}^+(S^{n-6} \times \mathbb{R}\mathrm{P}^2 \times \mathbb{C}\mathrm{P}^2)$$

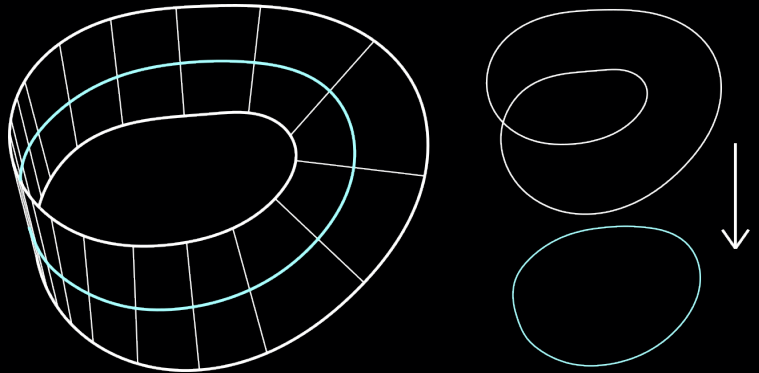
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$\rightarrow$ The manifold $S^{n-6} \times \mathbb{R}\mathrm{P}^2 \times \mathbb{C}\mathrm{P}^2$ is just an arbitrary reference manifold with the correct tangential 2-type.



θ-Structures and Tangential 2-Types

Scalar Curvature

Definition

Let (M, g) be a Riemannian manifold of dimension n. The scalar curvature of g is the function $\mathrm{scal}_g \colon M \to \mathbb{R}$ that appears in the taylor expansion

$$\frac{\mathrm{vol}(B_r(x))}{\mathrm{vol}(B_r^{\mathrm{euklid}})} = 1 - \frac{\mathrm{scal}_g(x)}{6(n+2)} r^2 + \dots$$

θ-Structures

Definition

Let $\theta\colon B \to \mathrm{BO}$ be a fibration. A θ-structure on a manifold M is a lift of the stable tangent bundle $\tau\colon M \to \mathrm{BO}$ along θ.

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$\to$ Generalizes orientations via $\theta\colon \mathrm{BSO} \to \mathrm{BO}$.

$\to$ Generalizes spin structures via $\theta\colon \mathrm{BSpin} \to \mathrm{BO}$.

$\to$ Every manifold has a canonical $\mathrm{id}_{\mathrm{BO}}$-structure.

Tangential Type

Definition

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$\rightarrow$ Unique up to fiber homotopy equivalence.

$\rightarrow$ Tangential 1-type is either $\mathrm{BSO} \to \mathrm{BO}$ or $\mathrm{id}_{\mathrm{BO}}$, depending on w_1.

Tangential 2-types

Theorem

Let $G = \pi_1(M)$ and w_i denote the Stiefel–Whitney-classes of M. Call $\widetilde{w}_2 = w_2(\widetilde{M})$. Then the tangential 2-type of M is:

- $\mathrm{BO} \times_{\mathrm{B}\mathbb{Z}_2} \mathrm{B}G$, if $\widetilde{w}_2 \neq 0, w_1 \neq 0$
- $\mathrm{BSO} \times \mathrm{B}G$, if $\widetilde{w}_2 \neq 0, w_1 = 0$
- $\mathrm{B}(G, w_1, w_2)$, if $\widetilde{w}_2 = 0$.

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$\rightarrow$ These are six cases, as $\widetilde{w}_2 \neq 0$ implies $w_2 \neq 0$.